

HRNova

Full HR System — Claude Code Development Prompts

14 Parts. One prompt per part. Build → Test → Next.

Item	Value
Total parts	14 — Foundation to Launch
Features	Employee Mgmt, Guard Mode Attendance, Leave, Payroll, Performance, AI Reports, Recruitment, Public Job Board, Multi-Branch, Notifications
Three apps	Web Dashboard, Guard Mode (Android), Employee Mobile (Android)
Company types	Single company + Multi-branch organizations
Rwanda compliance	2025 RSSB rates, PAYE brackets, RRA/RSSB Excel export
AI	OpenRouter — google/gemma-2-9b-it:free
Email	Brevo — free 9,000/month — fully cloud
Photos/CVs	Cloudflare R2 — zero egress fees
WhatsApp	Phase 2 — Green API (portal logic built now)

HOW TO USE THESE PROMPTS

1. Give **MASTER CONTEXT** first at the start of every new Claude Code session.
2. Give one part prompt at a time. Say: Build this.
3. Test every done condition at the bottom before moving to the next part.
4. Never skip a part. Never run two parts simultaneously.
5. If a part fails — fix it completely before continuing.

Part	Builds	Test On
Master Context	Full system context — give FIRST every session	—
1	Foundation: folders, theme, colors, router, widgets, backend scaffold	Chrome
2	Authentication: login screen, Firebase Auth, all roles, routing	Chrome
3	Super Admin: company management, multi-branch setup, billing	Chrome
4	Settings: onboarding wizard, company config, branch management	Chrome
5	Employees: profiles, profile photos, QR codes, loans, R2 upload	Chrome
6	Guard Mode: attendance scanning, all states, dashboard, calendar	Android phone
7	Leave: mobile app, web approvals, WhatsApp portal scaffold, email	Chrome + phone
8	Payroll: 2025 rates, payslips PDF, RRA/RSSB Excel export	Chrome
9	Performance: KPIs, scoring, AI reviews, charts, annual reports	Chrome
10	AI Reports: daily/weekly/monthly, group reports, Nova AI chat	Chrome
11	Recruitment: job board, public apply form, AI CV screening	Chrome + public URL
12	Mobile App: all 5 tabs, leave, payslip, boarding pass cards	Android phone
13	Multi-Branch: group dashboard, consolidated reports, branch filters	Chrome
14	Polish: final design, demo data, full testing, Firebase deployment	Chrome + phone

MASTER CONTEXT PROMPT

Give this to Claude Code at the start of EVERY new session before any part prompt.

► MASTER CONTEXT — Paste this FIRST in every Claude Code session

```
You are building HRNova – a full production-grade AI-powered HR Management
System built specifically for Rwanda. This is not an MVP. This is the complete,
full-featured HR system. Build production-quality code only.

=====
PRODUCT IDENTITY
=====
Name: HRNova
Tagline: Your HR Team, Supercharged
Market: Rwanda – factories, schools, clinics, NGOs, hotels,
        polytechnics, multi-branch institutions
Positioning: AI assistant that makes HR managers more powerful.
             NOT a replacement for HR staff.

=====
THREE APPS FROM ONE FLUTTER CODEBASE
=====
1. Web Dashboard – HR Admin, Manager, Director, Group HR Admin
2. Guard Mode App – Security guard smartphone at office gate
3. Employee Mobile App – All employees personal phones

=====
COMPANY TYPES – TWO STRUCTURES
=====
Type 1 – Single Company:
  One company, one HR Admin, all employees together.
  Example: factory, clinic, school, hotel.

Type 2 – Multi-Branch Organization:
  One parent organization with multiple branches.
  Each branch has its own Branch HR Admin.
  One Group HR Admin at top sees everything across all branches.
  Example: Rwanda Polytechnic with 6 campuses.
  Branch HR Admin sees only their branch.
  Group HR Admin sees all branches and can filter by branch.

=====
ROLES
=====
super_admin    – system owner. Sees all companies.
hr_admin       – full HR access for their company only.
group_hr_admin – multi-branch only. Sees all branches.
branch_hr_admin – multi-branch only. Sees only their branch.
finance_manager – expense claims and financial visibility.
manager        – department head. Leave approval, dept attendance.
guard          – Gate only. Guard Mode attendance scanning.
director       – read-only dashboard. Receives all AI reports.
employee       – mobile app only. Own records.

CRITICAL MULTI-TENANT RULE:
Every Firestore query MUST be scoped to companyId.
```

Company A data NEVER accessible to Company B.
Branch HR Admin queries MUST also be scoped to branchId.

COMPLETE MODULE LIST

1. Employee Management + Profile Photos + QR Codes
2. Attendance – Guard Mode (primary) + Manual entry
3. Leave Management – Mobile app + WhatsApp Leave Portal
4. Payroll – Full Rwanda 2025 rates + RRA/RSSB Excel export
5. Performance Management – KPIs + AI reviews + rankings
6. AI Reports – Daily, Weekly, Monthly via OpenRouter
7. Recruitment Panel – Job postings + AI CV screening + shortlisting
8. Public Job Board – Branded per company, no login needed
9. Notifications – Brevo email. WhatsApp via Green API (Phase 2).
10. Multi-Branch Reporting – Group HR consolidated reports
11. Super Admin Panel – Company management, billing, activation

TECHNOLOGY STACK – ALL FINAL DECISIONS

Frontend: Flutter (Dart) – web + android, one codebase
State: flutter_riverpod ^2.6.1
Navigation: go_router ^15.1.2
Backend: Node.js + Express – deployed to cloud (Railway/Render)
Database: Firebase Firestore
Auth: Firebase Auth + custom claims for all roles
Hosting: Firebase Hosting (Flutter web)
Photos/CVs: Cloudflare R2 – NEVER Firebase Storage
Access Key: e077e50e06c99119a933ea3a9285e539
Secret Key: a435aaf98d8b9be5078226718b3568dbeld61fdbd1bba568e369dec89b3e451f
Endpoint: https://66a88e290804df57603cf37cb27dc6ba.r2.cloudflarestorage.com
Bucket: hrnova-photos
AI: OpenRouter API – free model
Key: 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
Model: google/gemma-2-9b-it:free
Email: Brevo API – free 9,000/month – runs in cloud
Add BREVO_API_KEY to .env when account created
WhatsApp: Skip for now – add Green API in Phase 2
Excel: xlsx npm package – RRA/RSSB batch export
PDF: pdf Flutter package – payslips and reports
QR display: qr_flutter
QR scan: mobile_scanner
Camera: camera package
Image: image package – compress all photos to max 100KB
CV Parse: pdf-parse npm – extract text from applicant CVs
Charts: fl_chart Flutter package

FIREBASE PROJECT

Project ID: hrnova-6b7d8
Android App ID: 1:255829747858:android:105345226233101162de4b
Web App ID: 1:255829747858:web:df2e16b02011f02a62de4b
Client email: firebase-adminsdk-fbsvc@hrnova-6b7d8.iam.gserviceaccount.com

DESIGN SYSTEM – APPLE iOS 18 + macOS INSPIRED

PRIMARY: #4A9EFF light blue
ACCENT: #0A1628 deep navy
BACKGROUND: #F8FAFF soft white with blue tint
SIDEBAR: #0A1628 ALWAYS dark regardless of theme
SUCCESS: #1DB87A mint green
WARNING: #F5A623 warm amber
ERROR: #E5534B soft red
TEXT PRIMARY: #0A1628
TEXT SECONDARY: #6B7A99
CARD BORDER: 0.5px #E8EFF8
CARD RADIUS: 20px
CARD SHADOW: 0 2px 20px rgba(74,158,255,0.06)
BUTTON RADIUS: 10px
INPUT RADIUS: 10px
SIDEBAR WIDTH: 220px
TOPBAR HEIGHT: 64px

HUMAN-DESIGNED FEEL — CRITICAL:

Every screen must feel hand-crafted by a talented designer.
Not templated. Not generic. Warm, alive, intentional.
Employee avatar initials: unique gradient per person.
Metric cards: unique visual personality per metric type.
Empty states: personality and direction, not just no data.
Error messages: written like a helpful human, not a robot.
Leave balance cards: feel like boarding passes or tickets.

RWANDA 2025 TAX RATES — CRITICAL — USE EXACTLY

PAYE (monthly gross):

0 to 60,000 RWF:	0%
60,001 to 100,000 RWF:	20% on amount above 60,000
100,001 to 200,000 RWF:	8,000 + 30% on amount above 100,000
Above 200,000 RWF:	38,000 + 30% on amount above 200,000

RSSB (effective January 2025):

Pension employee:	6% of gross
Pension employer:	6% of gross
Maternity employee:	0.3% of gross
Maternity employer:	0.3% of gross
Occupational Hazard:	2% employer only

RSSB/PAYE deadline: 15th of following month

Maternity leave: 84 days — Rwanda law

Paternity leave: 4 days — Rwanda law

Rwanda public holidays:

01-01, 01-02, 02-01, 04-07, 05-01,
07-01, 07-04, 08-15, 12-25, 12-26

Currency: RWF — no decimals in display

Phone format: +250XXXXXXXXX

PHOTO AND STORAGE RULES — NEVER BREAK

ALL photos and files → Cloudflare R2 via Node.js backend

NEVER use Firebase Storage for anything

Compress all photos to max 100KB before upload

Delete attendance photos 2 weeks after monthly report sent

Store photoDeletionDate on every monthly report document

COMPLETE FIRESTORE STRUCTURE

```
companies/{companyId}/
  [name, industry, companyType (single|multi_branch),
   status, monthlyPrice, createdAt, rraTinNumber]
branches/{branchId}/
  [name, location, branchCode, branchHrAdminUid]
settings/config
  [workStartTime, workEndTime, gracePeriodMinutes,
   annualLeaveDays, sickLeaveDays,
   lateDeductionPerHourRwf, maxLateBeforeWarning,
   notificationMethod, isOnboardingComplete,
   workingDays[], departments[], performanceCriteria[],
   managerPhone, hrAdminPhone, guardPhone,
   managerEmail, hrAdminEmail,
   guardModeEnabled, whatsappLeavePortalEnabled,
   rraTinNumber, rraExportEnabled,
   salaryPaymentDay, overtimeMultiplier]
employees/{employeeId}/
  [firstName, lastName, nationalId, phone, email,
   dateOfBirth, emergencyContact,
   department, jobTitle, contractType,
   startDate, endDate, branchId,
   salaryType, salaryAmount, dailyRate, hourlyRate,
   transportAllowance, housingAllowance, bankAccount,
   role, qrCode, profilePhotoUrl, status,
   leaveBalances{annual,sick,maternity,paternity},
   loans[], createdAt]
attendance/{YYYY-MM-DD_employeeId}/
  [employeeId, employeeName, department, branchId,
   date, month, checkInTime, checkOutTime,
   checkInPhotoUrl, checkOutPhotoUrl,
   status, lateMinutes, totalHoursWorked,
   isApprovedLeave, isManualEntry,
   verificationType, guardUid, isGateGuard]
leave_requests/{requestId}/
  [employeeId, employeeName, department, branchId,
   leaveType, startDate, endDate,
   workingDaysRequested, reason, doctorNoteUrl,
   status, approvedBy, approvedAt, rejectionReason,
   source (mobile_app|whatsapp_portal|web),
   createdAt, month]
leaves_calendar/{YYYY-MM-DD_employeeId}/
  [employeeId, leaveType, approved, branchId]
payroll/{YYYY-MM}/
  [status, totalEmployees, totalGross, totalNet,
   totalPensionEmployee, totalPensionEmployer,
   totalMaternityEmployee, totalMaternityEmployer,
   totalOccupationalHazard, totalPaye,
   branchId (null for single companies),
   runBy, approvedBy, runAt, approvedAt,
   rraBatchExportedAt, photoDeletionDate]
payslips/{employeeId}/ [all payslip fields]
performance/{YYYY-MM_employeeId}/
  [employeeId, employeeName, department, branchId,
   month, scores{}, overallScore,
   aiReview, managerNotes, scoredBy, scoredAt]
```

```

job_postings/{jobId}/
  [title, department, description, requirements,
   minExperience, requiredSkills[], salaryMin,
   salaryMax, deadline, status,
   aiCriteria, publicUrl, totalApplications,
   shortlistedCount, createdAt]
applications/{applicationId}/
  [applicantName, email, phone, yearsExperience,
   coverLetter, cvUrl, cvKey,
   status (pending|shortlisted|declined|hired),
   aiScore, aiQualificationScore,
   aiExperienceScore, aiSkillsScore,
   aiCommunicationScore, aiRecommendation,
   aiStrengths[], aiConcerns[], aiSummary,
   interviewDate, interviewTime,
   rejectionConfirmedByHR, rejectionSentAt,
   appliedAt]
reports/{reportId}/
  [type, content, generatedAt, sentAt,
   branchId, photoDeletionDate, photosDeleted]
notifications/{notifId}/
  [userId, title, body, read, type, createdAt]
expense_claims/{claimId}/
  [employeeId, employeeName, department, branchId,
   expenseType, amount, date, description,
   receiptUrl, receiptKey, status,
   approvedBy, approvedAt, rejectionReason,
   reimbursedBy, reimbursedAt, month]
whatsapp_sessions/{phone}/
  [step, employeeId, employeeName, companyId,
   leaveType, startDate, endDate,
   workingDaysRequested, createdAt, expiresAt]
super_admin/companies_registry [all company summaries]

```

BACKEND FIREBASE SERVICE ACCOUNT

File: backend/serviceAccount.json

```

{
  type: service_account,
  project_id: hrnova-6b7d8,
  private_key_id: 75268fae3e27bf5e35b45e6a0f076038a68f9344,
  private_key: [the full private key from original setup],
  client_email: firebase-adminsdk-fbsvc@hrnova-6b7d8.iam.gserviceaccount.com,
  client_id: 114841821803875832897
}

```

Now wait for the specific part prompt. Do not build anything yet.

PART 1 — Project Foundation

Builds: folder structure, theme, colors, router, all shared widgets, backend Node.js scaffold

► PART 1 — PROJECT FOUNDATION

HRNOVA — PART 1 OF 14: PROJECT FOUNDATION

Build only what is listed. Do not add features not listed here.

CONTEXT: Flutter project hrnova exists. Firebase connected.
firebase_options.dart exists. pubspec.yaml exists.

TASK 1 — pubspec.yaml dependencies — add all:

```
firebase_core: ^3.13.1
firebase_auth: ^5.5.3
cloud_firestore: ^5.6.9
flutter_riverpod: ^2.6.1
go_router: ^15.1.2
qr_flutter: ^4.1.0
mobile_scanner: ^7.0.1
camera: ^0.11.1
image_picker: ^1.1.2
image: ^4.3.0
dio: ^5.8.0
pdf: ^3.11.3
printing: ^5.14.2
fl_chart: ^0.70.0
intl: ^0.20.2
shared_preferences: ^2.5.3
flutter_local_notifications: ^18.0.1
```

Run: flutter pub get

TASK 2 — Folder structure inside lib/:

```
core/
  theme/ app_colors.dart app_theme.dart
  constants/ app_constants.dart
  router/ app_router.dart
  services/ firebase_service.dart api_service.dart
features/
  auth/screens/login_screen.dart
    providers/auth_provider.dart
  dashboard/screens/dashboard_screen.dart
  employees/
    screens/ employees_screen.dart employee_profile_screen.dart
    models/ employee_model.dart
    providers/ employees_provider.dart
  attendance/
    screens/ attendance_screen.dart guard_mode_screen.dart
    models/ attendance_model.dart
  leave/
    screens/ leave_screen.dart
    models/ leave_request_model.dart
  payroll/
    screens/ payroll_screen.dart
    models/ payroll_model.dart
    services/ payroll_engine.dart
  performance/
    screens/ performance_screen.dart
```



```

models/ performance_model.dart
reports/
  screens/ reports_screen.dart nova_ai_screen.dart
recruitment/
  screens/ recruitment_screen.dart job_posting_screen.dart
  application_detail_screen.dart
models/ job_posting_model.dart application_model.dart
public/
  screens/ job_board_screen.dart apply_screen.dart
  application_success_screen.dart
settings/
  screens/ settings_screen.dart onboarding_screen.dart
models/ company_settings_model.dart
providers/ settings_provider.dart
super_admin/screens/ super_admin_screen.dart
mobile/screens/ mobile_home_screen.dart
branches/
  screens/ branches_screen.dart
  models/ branch_model.dart
shared/widgets/
  hrnova_button.dart
  hrnova_text_field.dart
  loading_widget.dart
  hrnova_sidebar.dart
  status_badge.dart
  employee_avatar.dart
  hrnova_card.dart
  metric_card.dart
main.dart

```

TASK 3 – app_colors.dart:

```

primaryBlue = Color(0xFF4A9EFF)
darkNavy = Color(0xFF0A1628)
backgroundBlue = Color(0xFFFF8FAFF)
successGreen = Color(0xFF1DB87A)
warningAmber = Color(0xFFFF5A623)
errorRed = Color(0xFFE5534B)
textPrimary = Color(0xFF0A1628)
textSecondary = Color(0xFF6B7A99)
cardBorder = Color(0xFFE8EFF8)
lightBlue50 = Color(0xFFFF0F6FF)
lightBlue100 = Color(0xFFEEF5FF)
pillGreenBg = Color(0xFFE8FBF3)
pillGreenText = Color(0xFF1DB87A)
pillAmberBg = Color(0xFFFFEF4E6)
pillAmberText = Color(0xFFFF5A623)
pillRedBg = Color(0xFFFFEEEE)
pillRedText = Color(0xFFE5534B)
pillBlueBg = Color(0xFFEEF5FF)
pillBlueText = Color(0xFF4A9EFF)
sidebarActive = Color(0x154A9EFF)

```

Avatar gradient colors per letter A-Z:

Create a static Map<String, List<Color>> avatarGradients

Each letter maps to a unique 2-color gradient.

Example: A → [Color(0xFF667EEA), Color(0xFF764BA2)]

B → [Color(0xFF11998E), Color(0xFF38EF7D)]

C → [Color(0xFFEB3349), Color(0xFFFF45C43)]

Make 26 unique, beautiful gradients – no two alike.

Used to give each employee avatar a unique feel.

TASK 4 – app_theme.dart:

```
lightTheme: background #F8FAFF, primary #4A9EFF,  
  cards white 20px radius 0.5px border #E8EFF8,  
  subtle shadow 0 2px 20px rgba(74,158,255,0.06)  
darkTheme: background #070E1C, cards #0D1628,  
  card border rgba(74,158,255,0.12)  
Sidebar ALWAYS #0A1628 regardless of theme  
Fonts: system default (no custom fonts needed)
```

TASK 5 – app_constants.dart:

```
appName = HRNova  
appTagline = Your HR Team, Supercharged  
backendBaseUrl = http://localhost:3000  
Roles: super_admin, hr_admin, group_hr_admin,  
  branch_hr_admin, finance_manager,  
  manager, guard, director, employee  
Company types: single, multi_branch  
Leave types: annual, sick, maternity, paternity, unpaid, emergency  
Leave sources: mobile_app, whatsapp_portal, web  
Contract types: permanent, fixed_term, probation  
Salary types: fixed_monthly, daily_rate, hourly_rate  
Verification types: guard_mode, manual, selfie  
Expense types: transport, fuel, meals, accommodation, entertainment, other  
Application statuses: pending, shortlisted, declined, hired  
Rwanda holidays: ['01-01','01-02','02-01','04-07','05-01',  
  '07-01','07-04','08-15','12-25','12-26']  
RWANDA 2025 TAX RATES as double constants:  
  pensionEmployeeRate = 0.06  
  pensionEmployerRate = 0.06  
  maternityEmployeeRate = 0.003  
  maternityEmployerRate = 0.003  
  occupationalHazardRate = 0.02  
  payeTaxFreeMonthly = 60000.0  
  payeBracket1Max = 100000.0  
  payeBracket1Rate = 0.20  
  payeBracket2Max = 200000.0  
  payeBracket2Rate = 0.30  
  payeBracket3Rate = 0.30  
  rraDeadlineDay = 15  
  maternityLeaveDays = 84  
  paternityLeaveDays = 4
```

TASK 6 – app_router.dart using GoRouter:

```
Routes:  
  /login → LoginScreen  
  /dashboard → DashboardScreen  
  /employees → EmployeesScreen  
  /employees/:id → EmployeeProfileScreen  
  /attendance → AttendanceScreen  
  /guard-mode → GuardModeScreen  
  /leave → LeaveScreen  
  /payroll → PayrollScreen  
  /performance → PerformanceScreen  
  /reports → ReportsScreen  
  /nova-ai → NovaAiScreen  
  /recruitment → RecruitmentScreen  
  /recruitment/new → JobPostingScreen
```

```
/recruitment/:jobId → JobPostingScreen
/recruitment/:jobId/application/:appId → ApplicationDetailScreen
/settings → SettingsScreen
/super-admin → SuperAdminScreen
/mobile-home → MobileHomeScreen
/branches → BranchesScreen
/apply/:companySlug/:jobSlug → ApplyScreen (PUBLIC – no auth)
/jobs/:companySlug → JobBoardScreen (PUBLIC – no auth)
/apply-success → ApplicationSuccessScreen (PUBLIC)
Default: /login
/apply/* and /jobs/* routes must NOT require authentication.
All other routes require authentication.
```

TASK 7 – Every screen: placeholder showing screen name.

GuardModeScreen placeholder: dark background, shield icon, Guard Mode text.

Public screens: light clean layout, HRNova logo, coming soon text.

TASK 8 – Shared widgets:

HRNovaButton:

```
label, onPressed, isLoading, isFullWidth
backgroundColor (default primaryBlue), textColor white
Height 44px, radius 10px, 16px padding
Primary: subtle inner gradient for premium feel
Loading: CircularProgressIndicator white
```

HRNovaTextField:

```
label above field 12px medium
Background #F8FAFF, border 0.5px #D8E4F0
Focus: primaryBlue border with subtle glow
Height 44px, radius 10px
obscureText with show/hide toggle
```

LoadingWidget: centered CircularProgressIndicator primaryBlue
Optional message below. Beautiful not just a spinner.

StatusBadge:

```
text + type parameter
Types and colors from app_colors.dart pill colors
Radius 100px, 11px medium, padding 4px 10px
```

EmployeeAvatar:

```
Parameters: firstName, lastName, photoUrl, size
If photoUrl: circular network image
If no photo: circular gradient using avatarGradients
              based on first letter of firstName
Subtle ring in primaryBlue 1.5px
```

HRNovaCard:

```
child, padding (default 20px), title optional
White background, 20px radius, 0.5px border, soft shadow
```

MetricCard:

```
value (String), label, icon, trend optional, color optional
Large number 32px bold, label 12px uppercase letter-spacing
Icon in colored circle 36px
Trend pill if provided
```

HRNovaSidebar:
 ALWAYS #0A1628 background
 Top: HRNova logo (HR white, Nova #4A9EFF)
 Thin colored line below logo
 Company name 12px muted below
 Nav items: icon + label, section dividers
 Active: #4A9EFF15 bg, 3px left border #4A9EFF, text #4A9EFF
 Inactive: white 55% opacity
 Bottom: EmployeeAvatar small + name + role + green online dot
 Width: 220px fixed
 onItemTapped callback
 Roles control which nav items are visible

TASK 9 – firebase_service.dart:
 Static getters: db, auth, currentUserId
 getCurrentUserClaims() → Future<Map<String,dynamic>?>
 CollectionReference helpers:
 employeesRef(companyId)
 attendanceRef(companyId)
 leaveRef(companyId)
 payrollRef(companyId)
 performanceRef(companyId)
 reportsRef(companyId)
 jobPostingsRef(companyId)
 applicationsRef(companyId, jobId)
 settingsRef(companyId) → DocumentReference
 branchesRef(companyId)

TASK 10 – main.dart:
 Firebase.initializeApp with DefaultFirebaseOptions
 ProviderScope wrapping HRNovaApp
 MaterialApp.router with GoRouter
 Both light and dark ThemeData
 themeMode from shared_preferences provider
 title: HRNova, debugShowCheckedModeBanner: false

TASK 11 – Backend foundation (backend/ folder):

package.json with all dependencies:
 express, firebase-admin, cors, dotenv,
 @aws-sdk/client-s3, multer, axios,
 @getbrevo/brevo, node-cron, xlsx,
 pdf-parse, bcryptjs

 .env file with all placeholders:
 PORT=3000
 FIREBASE_SERVICE_ACCOUNT_PATH=./serviceAccount.json
 CLOUDFLARE_R2_ACCESS_KEY_ID=e077e50e06c99119a933ea3a9285e539

CLOUDFLARE_R2_SECRET_ACCESS_KEY=a435aaf98d8b9be5078226718b3568dbe1d61fdbd1bba568e369dec89b3e451f

CLOUDFLARE_R2_BUCKET_NAME=hrnova-photos

CLOUDFLARE_R2_ENDPOINT=https://66a88e290804df57603cf37cb27dc6ba.r2.cloudflarestorage.com

CLOUDFLARE_R2_PUBLIC_URL=https://66a88e290804df57603cf37cb27dc6ba.r2.cloudflarestorage.com/hrnova-photos

OPENROUTER_API_KEY=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
 OPENROUTER_MODEL=google/gemma-2-9b-it:free
 BREVO_API_KEY=ADD_WHEN_ACCOUNT_CREATED

```
BREVO_SENDER_EMAIL=noreply@hrnova.rw  
BREVO_SENDER_NAME=HRNova
```

```
server.js: Express port 3000, CORS, JSON body parser  
  Firebase Admin SDK init  
  Import all routes  
  GET /api/health → { status: ok, timestamp }
```

```
middleware/verifyToken.js:  
  Verify Firebase Bearer token  
  Attach decoded to req.user  
  Return 401 if invalid
```

```
services/ empty scaffold files:  
  storageService.js, aiService.js,  
  emailService.js, dataProcessor.js,  
  payrollEngine.js, cvParserService.js,  
  aiScreeningService.js
```

```
routes/ empty scaffold files with health placeholder:  
  auth.js, companies.js, employees.js,  
  attendance.js, leave.js, payroll.js,  
  performance.js, reports.js, recruitment.js,  
  storage.js, notifications.js, exports.js, branches.js
```

DONE CONDITIONS:

```
[ ] flutter pub get - zero errors  
[ ] flutter analyze - zero errors  
[ ] flutter run -d chrome - launches, shows login placeholder  
[ ] /guard-mode route navigates without crash  
[ ] /apply/test/test route works WITHOUT login  
[ ] All other routes require authentication  
[ ] EmployeeAvatar shows gradient when no photo  
[ ] MetricCard renders with large number and label  
[ ] HRNovaSidebar renders in darkNavy in both themes  
[ ] node server.js starts without errors  
[ ] GET /api/health returns { status: ok }  
[ ] flutter build web - no errors
```

PART 2 — Authentication and Role-Based Routing

Builds: login screen, Firebase Auth, custom claims, all 9 roles, routing, backend auth routes

► PART 2 — AUTHENTICATION

HRNOVA — PART 2 OF 14: AUTHENTICATION AND ROLE-BASED ROUTING

Part 1 complete. All placeholders and shared widgets exist.

TASK 1 — Login screen (replace placeholder):

Full-screen layout. Background: #0A1628 with very subtle radial gradient #4A9EFF at 5% opacity in top-right corner.
Center card 420px wide on web, full width on mobile:
Card: white, 24px radius, subtle shadow
HRNova logo: HR white 40px bold, Nova #4A9EFF 40px bold
Tagline: Your HR Team, Supercharged — #6B7A99 14px italic
Thin divider 0.5px #E8EFF8 with generous spacing
Email HRNovaTextField
Password HRNovaTextField with show/hide
Sign In HRNovaButton fullWidth
Error message: #E5534B 13px — sounds human not robotic
Loading state on button while signing in
Add a small subtle creative touch below the card:
Powered by HRNova — 11px #6B7A99

TASK 2 — auth_provider.dart:

authStateProvider: StreamProvider on authStateChanges
authNotifierProvider: AsyncNotifier with:
 signIn(email, password): friendly error messages
 signOut()
 getUserRole(): from ID token claims
 getCompanyId(): from claims
 getBranchId(): from claims (for branch HR Admin)
 getCompanyType(): from claims (single|multi_branch)
 getUserClaims(): full claims map
themeNotifierProvider: StateNotifier storing ThemeMode
 Persists in shared_preferences
 Toggle between light and dark

TASK 3 — Role-based routing:

RouterNotifier watching authStateProvider
Redirect after login by role:
 super_admin → /super-admin
 hr_admin → /dashboard (check onboarding first)
 group_hr_admin → /dashboard
 branch_hr_admin → /dashboard
 finance_manager → /dashboard
 manager → /dashboard
 director → /dashboard
 guard → /guard-mode
 employee → /mobile-home
 unknown → sign out + error snackbar
Not authenticated → /login
/apply/* and /jobs/* → ALWAYS accessible without auth
LoadingWidget while checking auth state

TASK 4 — Suspension check:

After login for hr_admin and all company roles:

```
Check companies/{companyId} status == suspended
If yes: full-screen suspension page
  HRNova logo, soft red warning icon
  Account Suspended – 20px semibold
  Your company account has been suspended.
  Please contact HRNova support: support@hrnova.rw
  Sign Out button only
```

TASK 5 – Onboarding check:

```
After login for hr_admin:
  Read settings/config isOnboardingComplete
  If false → redirect to /onboarding before /dashboard
  Add /onboarding route to router
```

TASK 6 – Backend auth routes (backend/routes/auth.js):

```
POST /api/auth/set-claims (super_admin only):
  Body: { uid, role, companyId, branchId?,
          companyType?, isSuperAdmin? }
  Sets Firebase Auth custom claims
  Returns { success: true }

POST /api/auth/create-user:
  Creates Firebase Auth user + sets custom claims
  Body: { email, password, role, companyId,
          branchId?, companyType?, displayName }
  Returns { uid, success: true }
```

TASK 7 – api_service.dart:

```
Dio base URL from AppConstants.backendBaseUrl
Auto-attach Firebase ID token to all requests
get, post, put, delete with error handling
```

CREATE FIRST SUPER ADMIN:

```
Firebase Console → Authentication → Add user manually
Email: admin@hrnova.rw Password: Admin2026!
Call POST /api/auth/set-claims:
{ uid: USER_UID, role: super_admin, isSuperAdmin: true }
```

DONE CONDITIONS:

```
[ ] Login screen looks premium with #0A1628 background
[ ] Creative touch visible – gradient, subtle detail
[ ] Wrong credentials show friendly human error message
[ ] Each role redirects to correct route
[ ] Chrome refresh keeps user logged in
[ ] Sign out returns to login
[ ] Guard role → /guard-mode
[ ] /apply/test/job works without login
[ ] node server.js starts no errors
[ ] POST /api/auth/create-user works
[ ] flutter analyze zero errors
```

PART 3 — Super Admin Panel and Company Management

Builds: company creation, both company types, billing, branch setup

► PART 3 — SUPER ADMIN PANEL

HRNOVA — PART 3 OF 14: SUPER ADMIN PANEL AND COMPANY MANAGEMENT

Parts 1-2 complete. Auth and routing work.

TASK 1 — Super admin screen (replace placeholder):

Dark theme throughout — #0A1628 background

Sidebar: HRNova logo, Companies, Billing, Sign Out

Top bar: Super Admin badge #4A9EFF, sign out button

Summary row at top:

MetricCards: Total companies, Active, Suspended, Monthly revenue

Companies table:

Columns: Name, Type, Industry, Status, Price, Created, Actions

Type badge: Single or Multi-Branch

Actions: View, Suspend/Activate

Search by name

+ Add Company button

TASK 2 — Add Company slide-out panel 480px from right:

Company name (required)

Company type: Single Location / Multi-Branch Organization

Industry: Factory, School, Clinic, NGO, Hotel,

Polytechnic, Bank, Church, Construction, Other

Address, Contact person name

HR Admin email (required), HR Admin phone

Number of employees, Monthly price RWF

TIN number (for RRA exports)

Temporary password (required, min 8 chars)

If Multi-Branch: show add first branch section:

Branch name, Branch location, Branch code

SAVE flow:

1. Create companies/{autoId} document
companyType: single or multi_branch
2. POST /api/auth/create-user for hr_admin
If multi_branch: role = group_hr_admin
3. If multi_branch: create first branch document
Create branch_hr_admin account for first branch
4. Show success dialog with all credentials

TASK 3 — Company detail view:

All company info displayed

Edit any field

Activate/Suspend toggle with confirmation

Payment history: date, amount, reference, method

Add Payment button

If multi_branch: list of branches with

branch name, employees, Branch HR Admin email

Add Branch button

TASK 4 — Backend (backend/routes/companies.js):

POST /api/companies/create


```
GET /api/companies/:id
PUT /api/companies/:id
PUT /api/companies/:id/status
POST /api/companies/:id/payment
POST /api/companies/:id/branches
GET /api/companies/:id/branches
```

TASK 5 – Billing view:

Monthly revenue total
Per-company payment status
Unpaid this month highlighted amber

DONE CONDITIONS:

- [] Super admin sees companies list with type badges
- [] Single company created end to end
- [] Multi-branch company created with first branch
- [] HR Admin or Group HR Admin logs in immediately
- [] Suspend and reactivate work
- [] Payment records add correctly
- [] Billing shows correct revenue totals

PART 4 — Company Settings, Onboarding Wizard, and Branches

Builds: 5-step onboarding wizard, settings screen, branch management, company settings model

► PART 4 — SETTINGS AND ONBOARDING

HRNOVA — PART 4 OF 14: COMPANY SETTINGS, ONBOARDING, AND BRANCHES
Parts 1-3 complete.

TASK 1 — CompanySettings model (company_settings_model.dart):

All fields with types and defaults:

```
workStartTime: String 08:00
workEndTime: String 17:00
gracePeriodMinutes: int 10
annualLeaveDays: int 18
sickLeaveDays: int 10
lateDeductionPerHourRwf: int 500
maxLateBeforeWarning: int 3
salaryPaymentDay: int 28
overtimeMultiplier: double 1.5
transportAllowanceRwf: int 0
housingAllowanceRwf: int 0
notificationMethod: String both
isOnboardingComplete: bool false
workingDays: List<String> [mon-fri]
departments: List<String> []
performanceCriteria: List<Map> []
managerPhone: String empty
hrAdminPhone: String empty
guardPhone: String empty
managerEmail: String empty
hrAdminEmail: String empty
guardModeEnabled: bool true
rraTinNumber: String empty
rraExportEnabled: bool true
fromFirestore(), toFirestore(), copyWith()
```

TASK 2 — settings_provider.dart:

```
settingsProvider: StreamProvider<CompanySettings?>
  Real-time stream from companies/{companyId}/settings/config
settingsNotifierProvider: StateNotifier
  updateSettings(CompanySettings) → writes to Firestore
```

TASK 3 — Onboarding wizard (5 steps, cannot be skipped):

Full-screen, clean white design
Progress bar top showing Step X of 5
Back and Next buttons

Step 1 — Work Schedule:

Work start time (HH:MM text input or time picker)
Work end time
Grace period in minutes
Working days checkboxes Mon-Sat

Step 2 — Leave Policy:

Annual leave days (default 18)
Sick leave days (default 10)

Info box: Maternity 84 days, Paternity 4 days
Fixed by Rwanda law – cannot change

Step 3 – Payroll Rules:

Salary payment day of month
Overtime multiplier (1x, 1.5x, 2x dropdown)
Late deduction per hour in RWF

Step 4 – Departments:

Text field + Add button
Departments shown as removable chips
At least one department required

Step 5 – Notifications and Contacts:

Manager WhatsApp number (+250)
HR Admin WhatsApp number (+250)
Guard phone number (+250)
Manager email, HR Admin email
Notification method: Email only (WhatsApp coming soon)
Company TIN number for RRA exports

Save: write to Firestore, set isOnboardingComplete: true
Redirect to /dashboard

TASK 4 – Settings screen:

Same sections as onboarding in scrollable page
Sections collapsible with headers
Save per section, success snackbar

TASK 5 – Branch model (branch_model.dart):

id, name, location, branchCode
branchHrAdminUid, branchHrAdminEmail
employeeCount, createdAt
fromFirestore(), toFirestore()

TASK 6 – Branches screen (for Group HR Admin and Super Admin):

List of all branches for the organization
Each branch: name, location, employee count, HR Admin
Add Branch button
Click branch → filter dashboard to that branch

DONE CONDITIONS:

- [] Onboarding wizard 5 steps navigate correctly
- [] Completing wizard saves all fields to Firestore
- [] isOnboardingComplete true after completion
- [] Returning HR Admin goes to dashboard not onboarding
- [] Settings screen shows and updates current values
- [] Departments list persists correctly
- [] Branches screen shows all branches for multi-branch orgs

PART 5 — Employee Management

Builds: employee profiles, profile photo upload to R2, QR codes, loans, employee list, profile screen

► PART 5 — EMPLOYEE MANAGEMENT

HRNOVA — PART 5 OF 14: EMPLOYEE MANAGEMENT

Parts 1-4 complete.

TASK 1 — Employee model (employee_model.dart):

```
id, firstName, lastName, nationalId, phone, email
dateOfBirth, emergencyContact
department, jobTitle, branchId
contractType (permanent|fixed_term|probation)
startDate, endDate (nullable)
salaryType (fixed_monthly|daily_rate|hourly_rate)
salaryAmount, dailyRate, hourlyRate (doubles)
transportAllowance, housingAllowance (doubles)
bankAccount, rssbNumber
role, qrCode, profilePhotoUrl, status (active|inactive)
leaveBalances: Map{annual, sick, maternity:84, paternity:4}
loans: List<Map{description,total,monthly,remaining,settled}>
createdAt
fromFirestore(), toFirestore(), fullName getter
```

TASK 2 — employees_provider.dart:

```
employeesProvider: StreamProvider<List<Employee>>
  Stream active employees, scoped to companyId
  If branch_hr_admin: also filter by branchId
employeeByQRProvider: FutureProvider.family on qrCode
employeeByIdProvider: FutureProvider.family on employeeId
```

TASK 3 — Employee list screen (replace placeholder):

```
HRNovaSidebar with correct nav for role
Top bar: Employees title, count badge, + Add Employee button
Summary row: total active, by department breakdown
Search: filter by name, nationalId, department
Filters: department dropdown, branch dropdown (multi-branch only),
        status (Active/Inactive), contract type
Table columns:
  EmployeeAvatar, Full name, Department, Job title,
  Contract type, Salary type, Branch (multi-branch only),
  Status badge, Actions
Actions: View, Edit, Deactivate
Click row → /employees/:id
```

TASK 4 — Add employee slide-out panel 480px from right:

4 sections with collapsible headers:

Personal Info:

```
firstName, lastName (required)
nationalId, phone (required, +250 format hint)
email, dateOfBirth picker, emergencyContact
```

Employment:

```
department dropdown from settingsProvider.departments
```

jobTitle (required)
If multi-branch: branch dropdown from branchesProvider
contractType dropdown
startDate picker
endDate picker (only if fixed_term)
rssbNumber (optional)

Profile Photo (critical for Guard Mode):

Upload from camera or gallery button
Preview circle thumbnail
Compress to 100KB using image package
Upload to R2: {companyId}/profiles/{autoId}.jpg
Show upload progress
Note shown: Required for Guard Mode visual verification

Salary:

salaryType dropdown
Show relevant field only:
 fixed_monthly → Monthly salary RWF
 daily_rate → Daily rate RWF
 hourly_rate → Hourly rate RWF
transportAllowance RWF
housingAllowance RWF
bankAccount

System Access:

role dropdown (employee, manager)
If manager: note that login will be created

SAVE flow:

Validate required fields
Auto-generate Firestore document ID
qrCode = {companyId}_{documentId}
profilePhotoUrl from R2 upload
Initialize leaveBalances from settingsProvider
Set status: active, createdAt: now
Save to Firestore
If role == manager: POST /api/auth/create-user
Show success with QR code preview

TASK 5 – Employee profile screen (/employees/:id):

Header: large EmployeeAvatar 80px, full name bold 22px,
 job title #6B7A99, department badge, status badge, Edit button
6 tabs: Profile | QR Code | Attendance | Leave | Payroll | Loans

Profile tab:

Personal, Employment, Salary sections, read-only

QR Code tab:

Large QR using qr_flutter – QR data = employee.qrCode
Company name above, employee name and ID below
Download as PNG button
Print button using printing package
Regenerate QR button with confirmation
QR looks like a professional ID badge

Attendance tab: placeholder 'Filled in Part 6'

Leave tab: placeholder 'Filled in Part 7'

Payroll tab: placeholder 'Filled in Part 8'

Loans tab:

List of all loans with progress bars

Each loan: description, total, monthly deduction,
remaining balance, estimated payoff date

Add Loan button:

Form: description, total amount, monthly deduction

Saves to employee.loans array

TASK 6 – Edit employee: same panel pre-filled, save updates Firestore

TASK 7 – Deactivate employee:

Confirmation dialog with employee name

Sets status: inactive

Does NOT delete record – history preserved

Inactive filter toggle on list

TASK 8 – Backend (backend/routes/employees.js):

GET /api/employees/:companyId

POST /api/employees/:companyId

PUT /api/employees/:companyId/:employeeId

PUT /api/employees/:companyId/:employeeId/status

GET /api/employees/:companyId/:employeeId/qr

TASK 9 – R2 storage service (backend/services/storageService.js):

S3Client with R2 credentials from .env

uploadFile(buffer, key, contentType) → public URL

deleteFile(key)

deleteFiles(keys[])

getPublicUrl(key)

TASK 10 – Storage route (backend/routes/storage.js):

POST /api/storage/upload-photo (verifyToken, multer.single)

companyId, employeeId, photoType in body

Key: {companyId}/{photoType}/{employeeId}_{timestamp}.jpg

Return { url, key }

DELETE /api/storage/delete-batch (verifyToken)

Body: { keys: [] }

Return { deleted: count }

DONE CONDITIONS:

- [] Add employee with all fields works
- [] Profile photo uploads to Cloudflare R2
- [] Photo visible in EmployeeAvatar with circular crop
- [] QR code generates and displays correctly
- [] QR code downloadable as PNG
- [] Employee list shows all with search and filters working
- [] Branch filter works for multi-branch organizations
- [] Employee profile 6 tabs all render
- [] Loan can be added to employee
- [] Edit saves correctly
- [] Deactivate hides from active list
- [] R2 upload route works – test with Postman

PART 6 — Guard Mode Attendance System

Builds: fullscreen guard scanner, all attendance states, attendance dashboard, calendar, manual entry

► PART 6 — GUARD MODE ATTENDANCE

HRNOVA — PART 6 OF 14: GUARD MODE ATTENDANCE SYSTEM

Parts 1-5 complete. Employees exist with QR codes and profile photos.

CRITICAL: Guard Mode is the PRIMARY attendance method.

CRITICAL: ALL photos to Cloudflare R2. NEVER Firebase Storage.

CRITICAL: Compress all photos to max 100KB.

TASK 1 — Guard Mode screen (guard_mode_screen.dart — full replacement):

Route: /guard-mode

Accessible: guard role AND hr_admin for testing

Full screen — NO sidebar — NO app bar

Background: #0A1628 dark always

TOP BAR (70px fixed):

Left: Shield icon #4A9EFF + Guard Mode text lightBlue bold 16px

Center: Company name white 14px

Right: Current time updating every second white bold 18px

Below: thin divider + Today: {n} checked in badge

n = StreamBuilder counting today attendance docs

CENTER: MobileScanner fills remaining screen

Continuous scanning, onDetect calls _onQRDetected

BOTTOM BAR (48px):

Center: Scan your employee ID card — white 55% opacity

STATES — full screen overlays:

LOADING (0.5 seconds):

Semi-dark overlay, white CircularProgressIndicator

Verifying... white 16px

NOT_FOUND (2 seconds then back to scanning):

Full #E5534B background

Large person-off icon white 80px

Employee Not Found — white bold 28px

Please contact HR Admin — white 16px

CHECK_IN_SUCCESS (3 seconds):

Full #1DB87A background

Large subtle checkmark watermark behind content

EmployeeAvatar 120px circle — show profile photo if exists

Employee full name — white bold 28px

Job title + department — white 16px

CHECK IN ✓ — white pill with darker green bg

07:58 AM — white 22px

ON TIME — white bold or LATE — amber background pill

LATE_ARRIVAL (3 seconds):

Full #F5A623 background

Same profile photo circle
 Employee name – white bold 28px
 LATE – {lateMinutes} minutes – white bold 22px
 Checked in at {time} – white 16px

CHECK_OUT_SUCCESS (3 seconds):
 Full #4A9EFF background
 Profile photo circle
 Employee name – white bold
 CHECK OUT ✓ – white
 Worked: {hours}h {minutes}m today – white 20px

ALREADY_DONE (2 seconds):
 Dark grey #2C2C2C background
 Profile photo
 Already completed today – white
 Checked in at {time} – grey

ON QR SCAN LOGIC:
 1. If not scanning state: return (prevent double scan)
 2. Pause scanner
 3. Show LOADING state
 4. Query employees where qrCode == scannedValue, limit(1)
 If empty: NOT_FOUND state, resume scanner after 2 seconds
 5. Get today date string: DateFormat('yyyy-MM-dd').format(now)
 6. Get attendance doc: attendance/{date}_{employeeId}
 7. Has checkOutTime: ALREADY_DONE state
 Has checkIn no checkOut: CHECK_OUT flow
 No doc: CHECK_IN flow

CHECK_IN FLOW:
 1. Load employee profilePhotoUrl
 2. Parse workStartTime from settingsProvider (HH:mm)
 3. Add gracePeriodMinutes to get deadline DateTime
 4. Compare DateTime.now() to deadline
 5. on_time or late + calculate lateMinutes
 6. Save to Firestore attendance/{date}_{employeeId}:
 employeeId, employeeName, department, branchId,
 date, month, checkInTime: Timestamp.now(),
 checkInPhotoUrl: null (guard verified visually),
 status, lateMinutes, isApprovedLeave: false,
 isManualEntry: false, recordedBy: currentUserId,
 verificationType: guard_mode, guardUid: currentUserId,
 isGateGuard: true
 7. Show appropriate state
 8. Resume scanner after 3 seconds

CHECK_OUT FLOW:
 1. Load profile photo
 2. Calculate totalHoursWorked from checkInTime to now
 3. Calculate earlyLeaveMinutes if before workEndTime
 4. Update Firestore doc:
 checkOutTime: Timestamp.now(),
 totalHoursWorked, earlyLeaveMinutes
 5. Show CHECK_OUT_SUCCESS state
 6. Resume scanner after 3 seconds

TASK 2 – Attendance dashboard (attendance_screen.dart – full replacement):

HRNovaSidebar

Top: Attendance title, date picker (default today)

If multi-branch: branch filter dropdown

Summary cards row:

- Present (green), Late (amber), Absent (red), On Leave (blue)
- All real-time StreamBuilders

Tabs: Today | History | Summary

TODAY tab:

StreamBuilder on attendance where date == today

If branch_hr_admin: also filter by branchId

Each row:

- EmployeeAvatar 40px with profile photo or initials gradient
- Full name bold + department small
- Check-in time, check-out or Still in office (amber)
- StatusBadge: on_time=green, late=amber, absent=red, leave=blue
- verificationType icon: shield for guard_mode, pencil for manual
- Late minutes shown if late

Filter: department dropdown, status dropdown

Manual Entry button (HR Admin only):

- Form: employee dropdown, date, check-in time, check-out time optional, reason
- Save with isManualEntry: true, verificationType: manual

HISTORY tab:

Date picker, same list for selected date

SUMMARY tab:

Month picker

Table: employee name, present days, late days, total late minutes, absent days, on leave days

Export as PDF button

TASK 3 – Employee profile Attendance tab (fill in placeholder):

Month selector

Summary: present, late, absent counts

Calendar view: colored circles per day

- Green = on time, Amber = late, Red = absent,
- Blue = approved leave, Grey = weekend/holiday

Detail table below calendar

TASK 4 – Dashboard metric cards – connect to real data:

Present Today: count attendance with status on_time or late

Late Arrivals: count with status late

On Leave: count leaves_calendar for today

Pending Leave: count leave_requests with status pending

Show last 5 today attendance records below cards

DONE CONDITIONS:

- [] Guard logs in and sees fullscreen QR scanner
- [] Invalid QR shows red NOT_FOUND state
- [] Valid QR shows employee profile photo
- [] On-time: full green screen with photo and name
- [] Late: full amber screen with minutes late
- [] Check-out: blue screen with hours worked
- [] Already done: grey screen

```
[ ] Guard mode counter updates live
[ ] Attendance saved with verificationType: guard_mode
[ ] HR Admin can access guard mode for testing
[ ] Attendance dashboard shows live data
[ ] Manual entry works for HR Admin
[ ] Dashboard metric cards show real numbers
[ ] Attendance tab in employee profile shows calendar
```

PART 7 — Leave Management

Builds: mobile leave request, web approvals, WhatsApp portal scaffold, Brevo email, notifications

► PART 7 — LEAVE MANAGEMENT

HRNOVA — PART 7 OF 14: LEAVE MANAGEMENT

Parts 1-6 complete. Attendance system working.

Leave submitted via THREE channels:

1. Employee mobile app (primary for smartphone users)
2. WhatsApp Leave Portal (for feature phone workers)
3. HR Admin web dashboard (on behalf of any employee)

TASK 1 — Leave request model (leave_request_model.dart):

All fields including branchId and source field

TASK 2 — Working days calculator (lib/core/services/working_days_service.dart):

static int calculate(start, end, workingDays, holidays)

Exclude days not in company workingDays list

Exclude Rwanda public holidays from AppConstants

Return exact count inclusive

TASK 3 — MOBILE APP leave request (mobile_home_screen.dart Leave tab):

Balance cards for each leave type:

Annual: {n}/{total} days — green progress bar

Sick: {n}/{total} days — blue progress bar

Maternity: 84 days — Rwanda law info card purple

Paternity: 4 days — Rwanda law info card teal

Design cards like boarding passes — tactile, human feel

Colored left border 4px, perforated detail if possible

Request Leave button → DraggableScrollableSheet:

Leave type dropdown with colored icons per type

Start date picker (min: today)

End date picker (cannot be before start)

Working days info box updates live in blue

Balance warning in amber if insufficient

Reason text field

Submit button → source: mobile_app

My Requests list:

Each: type badge, date range, days, status badge

Source badge: Via WhatsApp shown in blue for portal requests

Tap rejected: expand to see rejection reason

TASK 4 — WEB DASHBOARD leave screen (leave_screen.dart):

HRNovaSidebar

3 tabs: Pending Approvals | All Requests | Calendar

PENDING APPROVALS tab:

StreamBuilder where status == pending

If branch_hr_admin: also filter by branchId

Each card:

EmployeeAvatar + name bold + department small

Leave type badge with unique color per type:

annual: #4A9EFF, sick: #1DB87A, maternity: purple,
paternity: teal, unpaid: grey, emergency: #F5A623
Date range + working days count
Reason text (truncated, tap to expand)
Current balance shown for that leave type
Source badge if WhatsApp Portal (human touch)
Approve button (green) | Reject button (red)

APPROVE FLOW:

Check balance – if insufficient: warning dialog
Batch write:
1. Update leave_request status: approved
2. Deduct from employee leaveBalances
3. Write leaves_calendar per working day
 {employeeId, leaveType, approved: true, branchId}
4. Create notification for employee
Show success snackbar

REJECT FLOW:

Required rejection reason dialog
Update status: rejected with reason
Create notification for employee
Note: HR confirms – rejection message sent ONLY after confirmation

ALL REQUESTS tab:

Full history with filters: month, employee, type, status, branch
Export PDF button

CALENDAR tab:

Monthly calendar grid
Colored dots per day per employee on approved leave
Tap day: list of employees on leave that day with leave type

TASK 5 – WhatsApp Leave Portal backend scaffold:

backend/services/whatsappPortalService.js:
handleLeaveRequest(phone, messageText, sessionData):
Full conversational flow:
 Step 2: Ask leave type (1-4)
 Step 3: Ask start date (DD/MM/YYYY)
 Step 4: Ask end date or SAME
 Step 5: Confirm and create leave_request
 source: whatsapp_portal
Sessions in: whatsapp_sessions/{phone}
Note: WhatsApp connection added in Phase 2 with Green API
Build the logic now so it is ready when WhatsApp is added

TASK 6 – Attendance integration:

When building absent list: check leaves_calendar
Approved leave → On Leave blue, not Absent red
Update attendance dashboard accordingly

TASK 7 – Notifications (Firestore-based):

Bell in top bar reads notifications for currentUserId
Unread count red badge
Click: dropdown list of recent notifications
Each: title bold, body, time ago, read/unread dot
Tap: mark as read
Mark all read button
Notification bell also in mobile app top bar

TASK 8 – Employee profile Leave tab (fill placeholder):

- All balances with colored progress bars
- Leave history list with badges
- Tap rejected item: show rejection reason
- HR Admin can manually adjust balance with reason

TASK 9 – Email notifications via Brevo:

- Install @getbrevo/brevo in backend
- backend/services/emailService.js:
 - sendEmail(to, toName, subject, htmlContent)
 - sendLeaveApprovalEmail(employee, leaveRequest)
 - sendLeaveRejectionEmail(employee, leaveRequest)
- All emails: HRNova branded HTML template
 - Dark header #0A1628, HRNova logo, company name
 - White content body
 - Light footer Powered by HRNova
- Trigger email on leave approval and rejection

DONE CONDITIONS:

- [] Employee submits leave on mobile, appears in HR pending
- [] Working days calculated correctly excluding Rwanda holidays
- [] Approve deducts correct balance
- [] leaves_calendar written for each approved day
- [] Attendance shows On Leave not Absent for approved days
- [] Source badge shows Via WhatsApp for portal requests
- [] Reject with reason works
- [] HR confirms before rejection email sent
- [] Leave calendar shows colored dots
- [] Email sent on approval via Brevo
- [] Notification bell shows unread count
- [] Employee profile Leave tab shows balances and history
- [] Multi-branch: branch filter works on leave screen

PART 8 — Payroll Engine

Builds: correct 2025 Rwanda rates, payslip PDF, RRA PAYE Excel export, RSSB Excel export

► PART 8 — PAYROLL ENGINE

HRNOVA — PART 8 OF 14: PAYROLL ENGINE WITH CORRECT 2025 RWANDA RATES

Parts 1-7 complete.

RWANDA 2025 RATES — USE EXACTLY — EMBEDDED IN CODE AS CONSTANTS:

PAYE:

0 to 60,000 RWF: 0%
60,001 to 100,000: 20% on amount above 60,000
100,001 to 200,000: 8,000 + 30% on amount above 100,000
Above 200,000: 38,000 + 30% on amount above 200,000
PAYE taxable = adjustedGross (RSSB does NOT reduce PAYE base)

RSSB (January 2025):

Pension employee: 6%
Pension employer: 6%
Maternity employee: 0.3%
Maternity employer: 0.3%
Occupational Hazard employer only: 2%
Deadline: 15th of following month

TASK 1 — PayrollEngine service (payroll_engine.dart — pure Dart):

calculatePayslip(employee, attendanceList, leaveList,
settings, month) → PayslipData

Step 1 — Base salary:

fixed_monthly: baseSalary = salaryAmount
daily_rate: presentDays × dailyRate
hourly_rate: totalHoursWorked × hourlyRate

Step 2 — Gross:

gross = base + transportAllowance + housingAllowance

Step 3 — Unauthorized absent deduction:

absentDays = days with no attendance AND no approved leave
For fixed_monthly: deduction = (salary/workingDays) × absentDays
For daily/hourly: already excluded in Step 1

Step 4 — Late deduction:

totalLateMinutes = sum all lateMinutes from attendance
lateDeduction = (totalLateMinutes/60) × lateDeductionPerHourRwf

Step 5 — Adjusted gross:

adjustedGross = gross - absentDeduction - lateDeduction

Step 6 — RSSB employee deductions:

pensionEmployee = adjustedGross × 0.06
maternityEmployee = adjustedGross × 0.003
totalEmployeeRSSB = pensionEmployee + maternityEmployee

Step 7 — PAYE:

taxableIncome = adjustedGross (NOT reduced by RSSB)
Apply progressive brackets from AppConstants

Step 8 – Loan deductions:

- For each active loan: deduct monthlyDeduction
- Update remainingBalance
- If remaining <= 0: mark settled

Step 9 – Net salary:

- net = adjustedGross - totalEmployeeRSSB - paye - loanDeductions

Step 10 – Employer costs (recorded, not deducted from employee):

- pensionEmployer = adjustedGross × 0.06
- maternityEmployer = adjustedGross × 0.003
- occupationalHazard = adjustedGross × 0.02

Return PayslipData with ALL fields

TASK 2 – Payroll screen (replace placeholder):

- Month selector at top
- Branch filter (multi-branch: can run per branch or all)
- Check if payroll already exists for selected month
- RUN PAYROLL button
- Progress: Calculating employee X of N...
- After calculation:
 - Summary MetricCards: employees, gross, net, RSSB, PAYE
 - Preview table: name, dept, gross, deductions breakdown, net
 - Anomaly warnings amber: salary changed >20% from last month
 - Edit Adjustment per employee: add bonus, deduction, note
- APPROVE PAYROLL button → locks payroll
- After approval:
 - Send Payslips button
 - Export RRA PAYE button
 - Export RSSB button

TASK 3 – Payslip PDF (using pdf package):

- Professional layout:
 - Header: company name left, HRNova logo small right
 - Employee: name, ID, dept, job title, month/year
- EARNINGS table:
 - Basic salary, Transport, Housing, Overtime, Bonuses
 - Total Gross
- DEDUCTIONS table:
 - Absent deduction, Late deduction
 - Pension (6%), Maternity (0.3%), PAYE
 - Loan repayments, Total deductions
- Employer costs box (info only):
 - Pension 6%, Maternity 0.3%, Occ. Hazard 2%
- NET SALARY large bold
- Note: RSSB and PAYE due by 15th of following month
- Footer: Powered by HRNova | Generated {date}

TASK 4 – RRA PAYE Excel export:

- backend/services/rraExportService.js using xlsx package
- generateRRAPayExport(companyId, month):
 - Sheet: PAYE Declaration
 - Columns: Employee Name, National ID, Gross Salary, Transport Allowance, Total Taxable Income, Tax Free (60,000), Taxable Above Free, PAYE Amount
- Footer: totals, company TIN, declaration text

Return Excel buffer

```
generateRSSBExport(companyId, month):  
  Sheet: RSSB Declaration  
  Columns: Employee Name, National ID, RSSB Number,  
           Gross Salary, Pension Base,  
           Employee Pension 6%, Employer Pension 6%,  
           Employee Maternity 0.3%, Employer Maternity 0.3%,  
           Employer Occ Hazard 2%,  
           Total Employee Deductions,  
           Total Employer Contributions  
  Footer: totals, company TIN, declaration
```

```
backend/routes/exports.js:  
  GET /api/exports/rra-payee/:companyId/:month  
  GET /api/exports/rssb/:companyId/:month  
  Both return Excel file download
```

```
Flutter download buttons:  
  GET request to export endpoint  
  Trigger browser download on web  
  Save file on mobile
```

TASK 5 – Send payslips via email:

- After payroll approved, Send Payslips button
- Generate PDF per employee
- emailService.sendPayslipEmail(employee, pdfBytes, month)
- Progress: Sending X of N...
- Log any failed sends

TASK 6 – Employee profile Payroll tab (fill placeholder):

- Month selector
- Payslip summary for selected month
- Download payslip PDF button
- Previous months list

DONE CONDITIONS:

- [] PAYE test: 50,000 → 0 tax
- [] PAYE test: 80,000 → 4,000 tax
- [] PAYE test: 150,000 → 23,000 tax
- [] PAYE test: 300,000 → 68,000 tax
- [] Pension 6% employee correct
- [] Pension 6% employer correct
- [] Maternity 0.3% each correct
- [] Occupational hazard 2% employer only
- [] Payslip PDF downloads with correct layout
- [] RRA PAYE Excel downloads with correct columns
- [] RSSB Excel downloads with correct columns
- [] Payroll locks after approval
- [] Loan deductions work correctly
- [] Payslip email sends via Brevo

PART 9 — Performance Management

Builds: KPI criteria, monthly scoring, AI review generation, performance charts, annual reports

► PART 9 — PERFORMANCE MANAGEMENT

HRNOVA — PART 9 OF 14: PERFORMANCE MANAGEMENT

Parts 1-8 complete.

TASK 1 — Performance criteria setup (in settings screen new section):

HR Admin defines company performance criteria

Default criteria pre-loaded:

Attendance and Punctuality (20%)

Quality of Work (25%)

Teamwork and Collaboration (20%)

Initiative and Innovation (20%)

Communication Skills (15%)

Add, edit, remove criteria

Each criterion: name + weight percentage

Weights must sum to 100% — validate and show error if not

Save to settings.performanceCriteria array

TASK 2 — Performance model (performance_model.dart):

id, employeeId, employeeName, department, branchId

month (YYYY-MM), scores{}, overallScore

aiReview, managerNotes, scoredBy, scoredAt

fromFirestore(), toFirestore()

TASK 3 — Monthly scoring (manager view in performance screen):

Month selector

List of employees in manager's department

If multi-branch: scoped to branch

Each row: EmployeeAvatar, name, dept, Score button,
last score with small star rating

Score panel (slide from right 480px):

Employee header: avatar, name, job title

For each criterion: name, weight, slider 1 to 5

1=Poor, 2=Below Average, 3=Average, 4=Good, 5=Excellent

Visual star or dot indicator per score

Weighted contribution shown small

Overall score: weighted average, large display

Manager notes text area

Generate AI Review button:

POST /api/ai/generate-review with scores and employee info

Loading state: Generating review...

AI review appears in editable text area

Manager can edit before saving

Save button: saves to Firestore performance collection

TASK 4 — Performance dashboard (HR Admin view):

Month selector

MetricCards: company avg score, top dept, lowest dept

Department breakdown bar chart using fl_chart

Top 5 performers: avatar, name, dept, score stars

Needs attention: score below 2.5 or declined >1 point

Not yet scored this month: list with reminder button
If multi-branch: branch comparison bar chart

TASK 5 – Employee profile Performance tab (fill placeholder):

Line chart (fl_chart) showing monthly scores over time
Trend indicator: arrow up (improving), flat, arrow down (declining)
Monthly review list:
 Month, overall score stars, AI review text, manager notes
 Expandable for criterion breakdown

TASK 6 – Annual performance report:

Generate for any employee for any year button
POST /api/reports/annual-performance
Gemma AI writes full narrative from all monthly scores
Shows: quarterly averages, best/worst month,
 trend analysis, attendance correlation,
 recommendation: Excellent/Good/Satisfactory/Needs Improvement
Download as PDF

TASK 7 – Backend AI routes (backend/routes/ai.js):

POST /api/ai/generate-review (verifyToken):
 Body: { employeeName, jobTitle, criteria, scores, overallScore }
 Build prompt for Gemma via OpenRouter
 Return { review: text }

POST /api/ai/annual-performance (verifyToken):
 Body: { employee, monthlyScores[], attendanceSummary }
 Return { narrative: text }

AI service (backend/services/aiService.js):
 callOpenRouter(prompt, maxTokens) → String
 generatePerformanceReview(data) → String
 generateAnnualReport(data) → String

DONE CONDITIONS:

- [] HR Admin can configure performance criteria with weights
- [] Weights must sum to 100% – error if not
- [] Manager can score employees with slider interface
- [] Weighted average calculates correctly
- [] AI review generates from OpenRouter
- [] Review appears in editable text area before save
- [] Performance dashboard shows company averages
- [] Declining performance flagged correctly
- [] Employee profile performance chart shows trend
- [] Annual report generates with AI narrative
- [] Multi-branch: branch comparison chart works

PART 10 — AI Reports and Nova AI

Builds: daily/weekly/monthly AI reports, group reports for multi-branch, Nova AI chat, photo deletion

► PART 10 — AI REPORTS AND NOVA AI

HRNOVA — PART 10 OF 14: AI REPORTS AND NOVA AI

Parts 1-9 complete.

CRITICAL — Pre-processing rule:

NEVER send raw Firestore documents to AI.

ALWAYS build summary object first, then send only summary.

TASK 1 — Data processor (backend/services/dataProcessor.js):

```
buildDailySummary(db, companyId, date, branchId=null):  
  Read attendance for date (filter by branchId if provided)  
  Read leaves_calendar for date  
  Calculate: totalActive, present, late, absent, onLeave,  
            attendanceRate, lateEmployees[], absentEmployees[],  
            departmentBreakdown[]  
  Return summary object
```

```
buildWeeklySummary(db, companyId, startDate, branchId=null):  
  7 days of daily summaries aggregated  
  Week-on-week comparison  
  Leave requests submitted this week  
  Return weekly summary
```

```
buildMonthlySummary(db, companyId, month, branchId=null):  
  Full month attendance aggregated  
  Leave summary by type  
  Payroll totals if approved  
  Performance average scores  
  Return comprehensive monthly summary
```

```
buildGroupSummary(db, companyId, reportType, date):  
  For multi-branch organizations  
  Runs per-branch summaries and combines  
  Adds branch comparison table  
  Returns group-level summary
```

TASK 2 — AI service (backend/services/aiService.js):

```
callOpenRouter(prompt, maxTokens=800):  
  POST to https://openrouter.ai/api/v1/chat/completions  
  Headers: Authorization Bearer + HTTP-Referer + X-Title  
  Body: model, messages, max_tokens  
  Timeout: 30 seconds  
  On error: return Report generation temporarily unavailable
```

```
generateReport(summaryData, reportType, companyName, isGroup=false):  
  Build prompt by type: daily/weekly/monthly/group  
  Daily prompt: professional morning attendance brief for HR manager  
  Weekly: weekly HR summary with trends and comparison  
  Monthly: comprehensive executive report for director  
  Group: organization-wide consolidated report with branch comparison
```

Return generated text

```
answerQuestion(question, contextData, companyName):  
  Build context prompt, return answer
```

TASK 3 – Reports routes (backend/routes/reports.js):

```
POST /api/reports/daily (verifyToken):  
  Body: { companyId, date?, branchId? }  
  Build summary → AI → save → return  
  Store: reports/{date}_daily_{branchId or all}
```

```
POST /api/reports/weekly (verifyToken):  
  Same pattern, store as {startDate}_weekly
```

```
POST /api/reports/monthly (verifyToken):  
  Same + store photoDeletionDate = now + 14 days  
  Store: reports/{YYYY-MM}_monthly
```

```
POST /api/reports/group-daily (verifyToken, group_hr_admin only):  
  Runs for all branches, combines, generates group AI report  
  Report format as described in Group HR Admin study:  
    OVERALL: N of N employees present (X%)  
    BY CAMPUS: each branch with emoji and percentage  
    ALERTS: AI identified concerns
```

```
POST /api/reports/ask (verifyToken):  
  Body: { companyId, question, branchId? }  
  Fetch relevant context, ask AI, return answer
```

TASK 4 – Reports screen (replace placeholder):

HRNovaSidebar

If group_hr_admin: extra Group tab at start

4 tabs: Daily | Weekly | Monthly | Ask Nova

(Group HR Admin gets 5 tabs: Group | Daily | Weekly | Monthly | Ask Nova)

DAILY tab:

- Latest daily report in formatted text card
- Generated at timestamp
- Generate Today's Report button
- Loading: Generating report... with AI animation
- Last 7 days list – click to load
- Branch selector if multi-branch

WEEKLY tab: latest weekly + generate + history

MONTHLY tab: latest monthly + generate + photoDeletionDate shown

GROUP tab (group_hr_admin only):

- Consolidated report across all branches
- Branch attendance comparison table
- Per-branch attendance rate bars
- Generate Group Daily Report button

ASK NOVA tab (nova_ai_screen.dart):

- Chat interface – feels like messaging
- User messages: right aligned, #4A9EFF background
- AI responses: left aligned, white card with subtle border
- HRNova N icon for AI messages
- Input field bottom with send button

Suggested questions chips when empty:
How many employees came in late today?
Which department has best attendance this month?
How many leave requests are pending?
Which branch has lowest performance this quarter? (multi-branch)
Who has not taken leave in 3 months?
Loading: animated dots while AI responds
Error: friendly human message if AI unavailable

TASK 5 – Auto-send reports via email:

When Generate button clicked:

Generate report → send via Brevo email to configured recipients
Show: Report generated and sent to {n} recipients

TASK 6 – Photo deletion automation:

node-cron in server.js: 0 0 * * * (midnight daily)
Query reports where photoDeletionDate <= now
AND photosDeleted != true
For each: call DELETE /api/storage/delete-batch
Update report: photosDeleted: true, photosDeletedAt: now

DONE CONDITIONS:

- [] Daily report generates professional AI text
- [] Report saved to Firestore correctly
- [] Weekly and monthly reports generate
- [] Monthly report stores photoDeletionDate
- [] Group report generates for multi-branch organizations
- [] Group report shows per-branch breakdown
- [] Ask Nova answers: How many employees present today?
- [] Ask Nova answers: How many leave requests pending?
- [] Reports sent via Brevo email on generation
- [] Photo deletion cron runs and marks photosDeleted

PART 11 — Recruitment Panel and Public Job Board

Builds: job postings, public job board, application form, AI CV screening, interview invitations

► PART 11 — RECRUITMENT PANEL

HRNOVA — PART 11 OF 14: RECRUITMENT PANEL AND PUBLIC JOB BOARD

Parts 1-10 complete.

ARCHITECTURE:

Internal HR panel: HR Admin creates jobs and reviews applications

Public job board: branded per company, no login needed

AI screening: OpenRouter reads CV and scores applicant

HR confirms before ANY rejection is sent

TASK 1 — Job posting model (job_posting_model.dart):

id, title, department, description, requirements
minExperience, requiredSkills[], salaryMin, salaryMax,
deadline, status (draft|open|closed), aiCriteria,
publicUrl, companySlug, jobSlug,
totalApplications, shortlistedCount, createdAt
fromFirestore(), toFirestore()

TASK 2 — Application model (application_model.dart):

id, jobId, jobTitle, applicantName, email, phone,
yearsExperience, coverLetter, cvUrl, cvKey,
status (pending|shortlisted|declined|hired),
aiScore (0-100), aiQualificationScore,
aiExperienceScore, aiSkillsScore, aiCommunicationScore,
aiRecommendation (accept|review|reject),
aiStrengths[], aiConcerns[], aiSummary,
interviewDate, interviewTime, interviewLocation,
rejectionConfirmedByHR (bool),
rejectionSentAt, interviewInviteSentAt, appliedAt
fromFirestore(), toFirestore()

TASK 3 — Recruitment screen (internal HR Admin view):

HRNovaSidebar

Top: Recruitment title, + Post New Job button

Summary: Open positions, Total applications, Shortlisted, Hired

Job postings list:

Each job card: title, department, status badge,
applications count, shortlisted count,
deadline, View Applications button, Edit button

Status: Draft (grey), Open (green), Closed (red)

TASK 4 — Job posting form (job_posting_screen.dart):

Title, Department dropdown, Status toggle

Job description (rich text area)

Requirements (add one by one, shown as chips)

Required skills (add one by one, chips)

Minimum years experience (number)

Salary range: min and max RWF (optional — show to applicants?)

Application deadline (date picker)

AI Screening criteria:

What should AI prioritize when screening?

Text area for HR to describe ideal candidate

Example: Must have degree in accounting, experience with

Rwanda tax law, speaks Kinyarwanda
Save as Draft / Publish buttons
On Publish:
 Generate companySlug from company name
 Generate jobSlug from job title
 publicUrl = hrnova-6b7d8.web.app/jobs/{companySlug}/{jobSlug}
 Save to Firestore
 Show public link to copy and share

TASK 5 – Applications view (per job):

Click View Applications on any job
Summary: Total, Pending AI review, Shortlisted, Declined
Sort by: AI Score (highest first), Date applied
Filter: status, score range

Each application card:

Applicant name + avatar initials gradient
AI Score 0-100 large and bold (color coded:
 80-100: green, 60-79: blue, 40-59: amber, below 40: red)
AI Recommendation badge: Strong Match, Review, Not Suitable
Key strengths: 3 bullet points from AI
Years experience, Date applied
Download CV button
Shortlist button (green) | Decline button (red)
Click card → application detail screen

TASK 6 – Application detail screen:

Full AI analysis displayed:
 Score breakdown: qualification, experience, skills, communication
 Score bars per category
 AI written summary paragraph
 Strengths list, Concerns list
Applicant info: name, email, phone, experience
Cover letter full text
Download CV button (loads from R2)
View CV button (PDF viewer in modal)
Actions:
 SHORTLIST: update status, ask send interview invitation?
 If yes: interview date picker, time picker, location field
 emailService.sendInterviewInvitationEmail(applicant, job, interview)
 DECLINE: update status, mark rejectionConfirmedByHR: true
 Show pending rejections queue – HR sends in bulk
 Send Rejections button: sends professional email via Brevo

TASK 7 – PUBLIC JOB BOARD (job_board_screen.dart – NO AUTH NEEDED):

Route: /jobs/:companySlug
Fetch company by slug from Firestore (public read)
Company branding: name, industry
List of open jobs:
 Job title, department, salary range if public,
 deadline, Apply Now button
Clean professional design – represents the company well
HRNova powered by footer

TASK 8 – PUBLIC APPLICATION FORM (apply_screen.dart – NO AUTH NEEDED):

Route: /apply/:companySlug/:jobSlug
Fetch job posting from Firestore (public read)
Show: job title, company name, description, requirements
Application form:

Full name (required)
Email (required)
Phone (+250 format required)
Years of experience (number required)
Cover letter text area (optional but encouraged)
Upload CV: PDF only, max 5MB
Upload to R2: {companyId}/cvs/{jobId}/{timestamp}_{name}.pdf
Upload certifications: optional, PDF or image
Submit Application button

On Submit:

1. Upload CV to R2 via POST /api/storage/upload-cv (no auth)
2. Save application to Firestore
status: pending, appliedAt: now
3. Trigger AI screening (backend processes async):
POST /api/recruitment/screen-application
4. Send confirmation email to applicant via Brevo:
Thank you for applying. We will review and respond within 5 days.
5. Show success screen (application_success_screen.dart)

TASK 9 – AI CV Screening (backend/services/aiScreeningService.js):

```
screenApplication(applicationId, companyId, jobId):  
  1. Get application from Firestore  
  2. Get job posting for requirements and aiCriteria  
  3. Download CV from R2  
  4. Extract text using pdf-parse  
  5. Build scoring prompt for OpenRouter:  
     You are an expert HR recruiter for a Rwandan company.  
     Score this applicant from 0 to 100.  
     Job requirements: {requirements}  
     AI criteria: {aiCriteria}  
     CV text: {extractedText}  
     Cover letter: {coverLetter}  
     Return ONLY valid JSON:  
     { totalScore, qualificationScore, experienceScore,  
       skillsScore, communicationScore,  
       recommendation (accept|review|reject),  
       strengths: [3 items],  
       concerns: [1-2 items],  
       aiSummary: 2-3 sentence professional summary }  
  6. Parse JSON response  
  7. Update application in Firestore with all AI scores  
  8. Create notification for HR Admin:  
     New application received for {jobTitle}  
     AI Score: {score}/100 – {recommendation}
```

TASK 10 – Backend recruitment routes (backend/routes/recruitment.js):

```
POST /api/recruitment/jobs (verifyToken)  
PUT /api/recruitment/jobs/:jobId (verifyToken)  
GET /api/recruitment/jobs/:companyId (verifyToken)  
GET /api/recruitment/applications/:companyId/:jobId (verifyToken)  
PUT /api/recruitment/applications/:appId/shortlist (verifyToken)  
PUT /api/recruitment/applications/:appId/decline (verifyToken)  
POST /api/recruitment/send-rejections/:jobId (verifyToken)  
POST /api/recruitment/screen-application (NO AUTH – called async)  
POST /api/storage/upload-cv (NO AUTH – public form upload)  
GET /api/public/jobs/:companySlug (NO AUTH)  
GET /api/public/job/:companySlug/:jobSlug (NO AUTH)
```


DONE CONDITIONS:

- [] HR Admin creates job posting and gets public URL
- [] Public job board loads at /jobs/{slug} without login
- [] Apply form works without login
- [] CV uploads to Cloudflare R2 successfully
- [] Confirmation email sent to applicant via Brevo
- [] AI screening runs and scores the CV
- [] AI scores visible in applications list
- [] Applications sorted by AI score highest first
- [] Click application shows full AI analysis
- [] Shortlist button works and triggers interview email
- [] Decline does NOT send email until HR clicks Send Rejections
- [] HR Admin notified of new applications

PART 12 — Employee Mobile App

Builds: all 5 mobile tabs, boarding pass leave cards, payslip, notifications, responsive web

► PART 12 — EMPLOYEE MOBILE APP

HRNOVA — PART 12 OF 14: EMPLOYEE MOBILE APP

Parts 1-11 complete.

TASK 1 — Mobile home screen (full replacement /mobile-home):

Employee role only

Bottom navigation 5 tabs:

Home (house), Attendance (clock), Leave (calendar),

Payslip (document), Profile (person)

HOME TAB:

Top bar: HRNova logo left, notification bell right

Greeting: Good morning/afternoon/evening, {firstName}!

Today status card (gradient #4A9EFF to #2E7DE8):

Checked in: checkmark + time in white

On approved leave: blue leaf icon + On approved leave

Not checked in: grey + Not checked in yet

Add creative element: subtle wave SVG in card background

Quick stats 2x2 grid cards:

Present this month, Late this month,

Annual leave left, Pending leave requests

Recent attendance: last 5 records

Each: date, check-in time, status badge, hours worked

Request Leave button fullWidth #4A9EFF

ATTENDANCE TAB:

Month selector

Summary: Present X | Late X | Absent X

Calendar view — circles per day:

Green = on time, Amber = late, Red = absent,

Blue = approved leave, Grey = weekend/holiday

Daily detail list below calendar:

Date, check-in, check-out, status badge, hours

LEAVE TAB:

Balance cards designed like boarding passes:

White card, colored left border 4px

Annual: blue border, {n} days remaining large

Sick: green border, {n} days remaining

Maternity: purple border, 84 days Rwanda law

Paternity: teal border, 4 days Rwanda law

Subtle perforated pattern if possible

Request Leave button → DraggableScrollableSheet

Same form as web version

Submit creates leave_request with source: mobile_app

My Requests list:

Type badge, dates, days, status badge

Via WhatsApp badge if whatsapp_portal source

Tap rejected: expand rejection reason

PAYSLIP TAB:

Month selector

Net salary large bold #0A1628

Earnings breakdown collapsible
Deductions breakdown collapsible:
 RSSB Pension 6%, Maternity 0.3%, PAYE, others
Download payslip PDF button
Previous months list

PROFILE TAB:
EmployeeAvatar large 80px circle
Full name bold 20px, job title 14px, department badge
Company name, member since date
Info cards: National ID, Phone, Email
Sign Out button

TASK 2 – Mobile notifications:

Notification bell in home top bar
StreamBuilder on notifications where `userId == currentUserId`
Unread count red badge
Tap: bottom sheet with list
Each: title bold, body, time ago, read/unread blue dot
Tap: mark as read

TASK 3 – Responsive web dashboard:

Screen > 1024px: sidebar always visible
Screen < 768px: no sidebar, hamburger → drawer
Screen 768-1024px: collapsed sidebar icons only
Tablet checkin: always fullscreen, no sidebar

TASK 4 – Guard Mode improvements:

Small menu icon top right:
 Dashboard, Sign Out
Keep screen always on (WakeLock if available)
Show date below time in top bar

DONE CONDITIONS:

- [] Employee home shows correct today status
- [] Attendance calendar shows correct colors
- [] Leave balances show as boarding pass style cards
- [] Leave request submits and appears in HR pending
- [] After HR approves: employee sees Approved status
- [] Via WhatsApp badge shows on portal requests
- [] Payslip shows correct net salary and breakdown
- [] Payslip PDF downloads from mobile
- [] Notification bell shows unread count and messages
- [] Profile shows correct employee info
- [] Sign out works
- [] Web dashboard collapses on small screens

PART 13 — Multi-Branch Consolidated Reports

Builds: group HR dashboard, branch comparison reports, consolidated payroll, all branch filters

► PART 13 — MULTI-BRANCH REPORTS

HRNOVA — PART 13 OF 14: MULTI-BRANCH CONSOLIDATED REPORTS AND GROUP HR
Parts 1-12 complete.

This part finalizes the multi-branch features.
Single-company organizations are unaffected.

TASK 1 — Group HR Admin dashboard:

- Same layout as HR Admin dashboard but with branch overview
- Top bar: Group HR Admin badge + organization name
- Branch selector: All Branches + each branch name
- When All Branches selected:
 - MetricCards show organization totals
 - Attendance list shows all branches with branch column
 - Performance chart shows branch comparison bars
- When specific branch selected:
 - Dashboard shows only that branch data
 - Same as single-company HR Admin view for that branch

TASK 2 — Group HR daily WhatsApp report format:

Backend generates this format for group_hr_admin:
Good morning. {OrgName} Daily Report — {date}
OVERALL: {n} of {total} employees present ({rate}%)
BY BRANCH:

- ✅ {Branch1}: {n}/{total} — {rate}%
- ⚠️ {Branch2}: {n}/{total} — {rate}% (below 90%)

[one line per branch with emoji indicator]

ALERTS:

[AI identified concerns across all branches]
Note: For now this is generated and shown in dashboard.
WhatsApp delivery added in Phase 2 with Green API.
Email delivery via Brevo works now.

TASK 3 — Branch comparison report (Group HR Admin):

Accessible from Reports screen Group tab
Side by side table of all branches:

- Branch name, Employees, Attendance %, Late count,
- Leave pending, Avg performance score, Payroll cost

Color coding: green good, amber watch, red concern
This month vs last month trend arrows
Download as PDF button
Generate Group Monthly Report button:

- AI writes consolidated narrative covering all branches
- Includes: best and worst performing branch,
- organization-wide trends,
- recommendations for management

TASK 4 — Branch-level payroll:

If group_hr_admin: can run payroll per branch
Branch selector before Run Payroll
RRA and RSSB exports can be:

Per branch (Branch HR Admin runs for their branch)
Consolidated (Group HR Admin runs for all branches combined)

TASK 5 – Branch filter on all main screens:

Attendance screen: branch filter dropdown (group_hr_admin)
Leave screen: branch filter dropdown (group_hr_admin)
Payroll screen: branch selector (group_hr_admin)
Performance screen: branch comparison (group_hr_admin)
Employees screen: branch filter already in Part 5

DONE CONDITIONS:

- [] Group HR Admin dashboard shows all branches combined
- [] Branch filter switches view to specific branch
- [] Branch comparison table renders with all metrics
- [] Group monthly report generates consolidated AI narrative
- [] Payroll runs per branch and consolidated
- [] RRA/RSSB exports available per branch and combined
- [] All screens have branch filter for group_hr_admin
- [] Branch HR Admin cannot see other branches

PART 14 — Dashboard Polish, Demo Data, Testing, Deployment

Builds: final design polish, demo companies, complete testing checklist, Firebase deployment

► PART 14 — POLISH AND LAUNCH

HRNOVA — PART 14 OF 14: DASHBOARD POLISH, DEMO DATA, TESTING, DEPLOYMENT

Parts 1-13 complete. All features working. Final polish and launch.

TASK 1 — Complete HR Admin dashboard final design:

Apply full HRNova design language throughout:

Sidebar:

- Very subtle micro-texture in #0A1628
- Logo with thin #4A9EFF line below
- All nav items with correct icons
- Badge counts on Leave and Attendance
- Smooth hover transitions

Topbar:

- Dark/light toggle with smooth AnimatedTheme
- Notification bell with count
- User avatar with online dot
- Greeting changes by time of day

Dashboard content:

- Row 1: 4 MetricCards with real-time data
 - Each card has unique visual personality
- Row 2 left (60%): Today attendance list
 - Profile photos or gradient avatars
 - Verify icon (shield=guard, pencil=manual)
- Row 2 right (40%): Department attendance bars (fl_chart)
 - Bars are rounded, colors from design system
- Row 3 left: Last 3 pending leave requests
 - Quick approve/reject buttons
- Row 3 right: Quick Actions card
 - Generate Daily Report, Add Employee,
 - View Pending Leave, Run Payroll (if end of month)
- Row 4: Nova AI alerts (3 latest anomalies)
 - Each alert: emoji + description + timestamp

TASK 2 — Dark/light mode toggle:

- Moon/sun icon in topbar
- themeNotifierProvider toggle
- Store in shared_preferences
- AnimatedTheme wrapper for smooth transition
- Sidebar ALWAYS stays #0A1628 regardless of mode
- All screens respect current mode

TASK 3 — Empty states throughout app:

- Every list: custom empty state with personality
- No employees: person+ icon + Add your first employee + button
- No attendance: clock icon + No one has checked in yet today
- No leave requests: calendar icon + All leave is up to date
- No applications: briefcase icon + Post a job to receive applications
- No reports: chart icon + Generate your first report
- All empty states: icon + friendly message + action button

TASK 4 — Error handling everywhere:

- Every network call: loading state + error catch
- Snackbar errors: friendly human messages not error codes

Connection error: offline banner with retry button
Auth error: auto sign out and redirect

TASK 5 – Demo data seed script (backend/scripts/seedDemo.js):

Run: node backend/scripts/seedDemo.js
Creates: Kigali Demo Factory (single company)
 industry: Factory, companyType: single
 monthlyPrice: 400000
 rraTinNumber: 123456789
Creates: Rwanda Polytechnic Demo (multi-branch)
 companyType: multi_branch
 Branches: RP Kigali, RP Huye, RP Musanze
 3 branch HR Admins + 1 group HR Admin
Creates settings with realistic Rwanda values
Creates 40 employees for factory (mix of salary types)
Creates 20 employees per RP branch (60 total)
All employees have realistic Rwandan names
Creates profile photos using placeholder URLs
Creates 90 days attendance (95% rate, realistic patterns)
Creates 15 leave requests (mix of approved/pending/rejected)
Creates 2 months payroll with correct 2025 rates
Creates performance scores for 3 months
Creates 3 open job postings with 10 applications each
Creates 5 AI reports

Demo accounts:

admin@hrnova.rw / Admin2026! – super_admin
demo@hrnova.rw / Demo2026! – hr_admin (factory)
rp@hrnova.rw / Demo2026! – group_hr_admin (RP)
guard@hrnova.rw / Demo2026! – guard (factory)
employee@hrnova.rw / Demo2026! – employee (factory)
finance@hrnova.rw / Demo2026! – finance_manager

TASK 6 – Final testing checklist:

AUTH AND ROLES:

- [] Super admin → /super-admin
- [] HR Admin → /dashboard or onboarding
- [] Group HR Admin → /dashboard with branch controls
- [] Guard → /guard-mode
- [] Employee → /mobile-home
- [] Wrong password shows friendly error
- [] Suspended company shows suspension screen

MULTI-BRANCH:

- [] Group HR Admin sees all branches combined
- [] Branch HR Admin sees only their branch
- [] Branch filter switches correctly
- [] Group consolidated report generates

GUARD MODE:

- [] QR scan identifies correct employee
- [] Employee profile photo shows in overlay
- [] On-time: green screen with photo and name
- [] Late: amber with minutes calculation correct
- [] Check-out: hours worked calculation correct
- [] Already done: grey screen prevents double entry

LEAVE:

- [] Mobile app leave request submits
- [] Working days exclude weekends and Rwanda holidays
- [] HR Admin approves and balance deducts
- [] leaves_calendar written for each approved day
- [] Attendance shows On Leave not Absent
- [] Rejection held until HR sends bulk rejections
- [] Email sent on approval via Brevo

PAYROLL:

- [] PAYE: 80,000 → 4,000 RWF
- [] PAYE: 150,000 → 23,000 RWF
- [] PAYE: 300,000 → 68,000 RWF
- [] Pension: exactly 6% employee and employer
- [] Maternity: exactly 0.3% each
- [] Occupational Hazard: 2% employer only
- [] Payslip PDF correct layout and figures
- [] RRA PAYE Excel downloads with correct columns
- [] RSSB Excel downloads with correct columns
- [] Payroll locks after approval
- [] Payslip emails send via Brevo

PERFORMANCE:

- [] Criteria weights sum to 100% – error if not
- [] AI review generates from OpenRouter
- [] Performance chart shows trend over time
- [] Annual report generates with AI narrative

RECRUITMENT:

- [] Job posting creates public URL
- [] Public job board loads without login
- [] Application form works without login
- [] CV uploads to Cloudflare R2
- [] Confirmation email sent to applicant
- [] AI screening scores the CV
- [] Applications sorted by AI score
- [] Shortlist triggers interview invitation email
- [] Decline held until HR sends bulk rejections

REPORTS:

- [] Daily report generates AI text
- [] Group report shows per-branch breakdown
- [] Nova AI answers attendance and leave questions
- [] Monthly report stores photoDeletionDate
- [] Photo deletion cron runs correctly
- [] Reports emailed via Brevo

MOBILE:

- [] All 5 tabs load with real data
- [] Leave balance cards look like boarding passes
- [] Leave request submits from mobile
- [] Payslip visible and downloadable
- [] Notifications show and mark as read

DEMO:

- [] demo@hrnova.rw logs in with realistic factory data
- [] rp@hrnova.rw sees all 3 RP branches
- [] guard@hrnova.rw goes to Guard Mode
- [] employee@hrnova.rw sees mobile app
- [] Public job board shows jobs at /jobs/kigali-demo-factory

TASK 7 – Production build and deployment:

```
flutter build web --release
firebase deploy --only hosting
Verify all features on Firebase Hosting URL
Update AppConstants.backendBaseUrl for production
```

DONE CONDITIONS – ALL MUST PASS:

```
[ ] All checklist items above pass
[ ] flutter analyze – zero errors
[ ] flutter build web – no errors
[ ] Demo accounts all work on production URL
[ ] HRNova is ready for first real paying client
```

QUICK REFERENCE

Demo Accounts

Email	Password	Role	Company
admin@hrnova.rw	Admin2026!	super_admin	All companies
demo@hrnova.rw	Demo2026!	hr_admin	Kigali Demo Factory
rp@hrnova.rw	Demo2026!	group_hr_admin	Rwanda Polytechnic Demo
guard@hrnova.rw	Demo2026!	guard	Kigali Demo Factory
employee@hrnova.rw	Demo2026!	employee	Kigali Demo Factory
finance@hrnova.rw	Demo2026!	finance_manager	Kigali Demo Factory

Key Commands

Command	What It Does
flutter run -d chrome	Run web app in Chrome browser
flutter build web --release	Build for production
firebase deploy --only hosting	Deploy to Firebase Hosting
cd backend && node server.js	Start Node.js backend
node backend/scripts/seedDemo.js	Create demo companies and data
flutter analyze	Check for Dart errors
flutter pub get	Install Flutter packages
npm install	Install Node.js packages (in backend/ folder)

Critical Rules — Never Break

1. **ALL photos and files → Cloudflare R2 via backend. NEVER Firebase Storage.**
2. Compress ALL photos to max 100KB before upload.
3. Delete attendance photos 2 weeks after monthly report. Store photoDeletionDate.
4. Every Firestore query scoped to companyId. Branch queries also scoped to branchId.
5. RSSB pension: 6% employee + 6% employer (January 2025). NOT 3% and 5%.
6. PAYE tax-free threshold: 60,000 RWF per month. NOT 30,000.
7. Maternity: 84 days. Paternity: 4 days. Rwanda law. Not editable.
8. PAYE taxable income = adjustedGross. RSSB does NOT reduce PAYE base.
9. HR confirms before rejection emails are sent. Never automatic.
10. Sidebar ALWAYS stays #0A1628 regardless of light or dark mode.
11. Design must feel human-crafted — warm, alive, not templated or generic.
12. Employee avatar initials always use unique gradient per person — never flat grey.

13. WhatsApp portal logic built now — Green API connection added in Phase 2.
14. Public routes /apply/* and /jobs/* must work WITHOUT authentication.
15. Payroll locked after approval — corrections as adjustments next month only.